



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam, Fall 2025

CSE 3811 : Artificial Intelligence

Total Marks: 30

Duration: 1 hour 30 Minutes

Answer all the questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. A city hospital uses an intelligent robot to deliver medicines from the pharmacy to patient rooms through corridors and intersections. Some corridors may be blocked, and the robot has a limited battery. The robot starts at the pharmacy, must deliver medicine to a specific patient room, and should avoid blocked corridors while minimizing battery usage. Each movement from one location to another consumes one unit of battery.
 - a) Write the **PEAS** description for the robot. [2]
 - b) Formulate this problem as a search problem. [4]

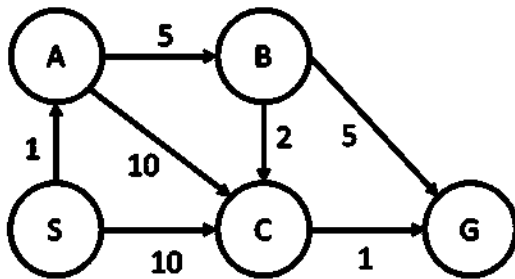


Figure 2(a)

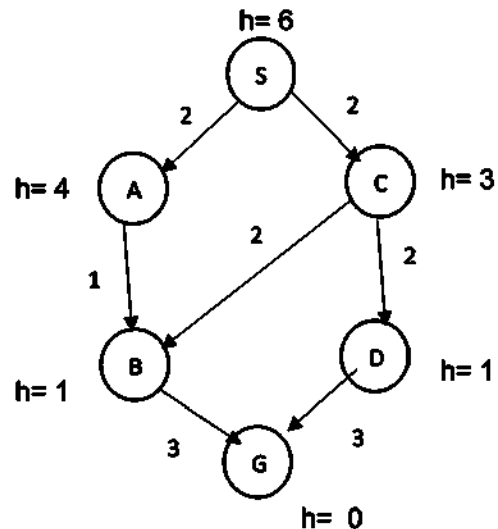


Figure 3(a)

2.
 - a) Show the corresponding tree representation of the following state space graph and apply the following search strategies showing the order of expansion, solution paths and path costs (where S is the start state and G is the goal state). [2+2]
 - i) **DFS**
 - ii) **UCS**
 - b) Suppose you are trying to solve a search problem with a search tree having **branching factor b**, **maximum depth m**, and **shallowest solution depth s**. The root is at level 0 and the search algorithms operate **from left to right**. Additionally, assume that expanding any node has **unit cost**, the search stops immediately upon finding the **first goal**. [1+1]

- i) Which search strategy is guaranteed to expand **the fewest nodes for the following two cases?**
- When the shallowest solution is at the leftmost node
 - When the shallowest solution is at the rightmost node
- ii) How would your answers change if the search tree had **infinite depth ($m = \infty$)** and solutions are guaranteed to exist?
3. Consider the graph at figure 3(a) as state space graph where S is the start state and G is the goal state. h represents the heuristic values of the states.
- Show the solution paths and the associated actual path costs for the following search algorithms: [2+2]
 - A* Tree Search**
 - A* Graph Search**
 - Will A* graph search provide an optimal solution for the given state-space graph? If not, what is the particular reason behind it, explain briefly. [2]
- 4.

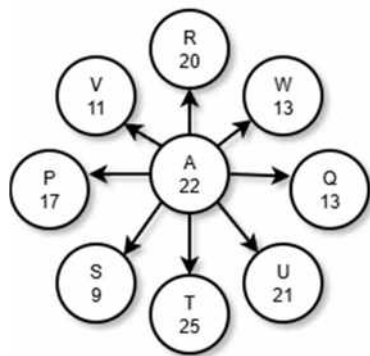


Figure 4(a)

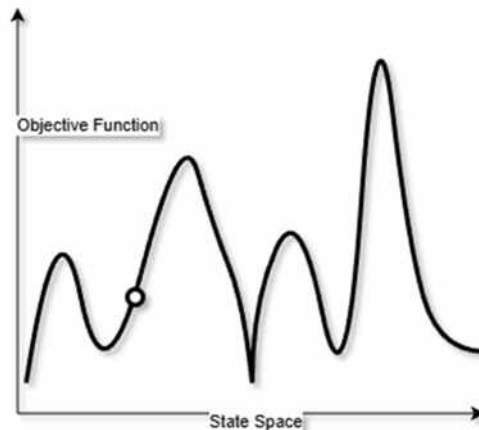


Figure 4(b)

- Santa Claus is currently at node A and wants to travel to a new node. He decides to use the simulated-annealing algorithm to pick his next node. He used an objective function to get the values for each node. [4]

The Temperature(T) at any time-step(t) is given by:

$$T = t - 2t^2 + 67$$

The Random Generator generates any number(N) between 0 to 1 by following the function:

$$N = |\cos(185 + 2.7t)|$$

Here cosine is in degrees and t is the time-step.

Starting from timestep $t=0$, one neighbor will be selected for calculation for each timestep (including $t=0$) according to the alphabetical order.

According to the algorithm, which node will he decide to **travel to first** and at what **timestep** will he make that decision? Show your calculations.

b) Determine which algorithm between stochastic-hill-climbing and simulated-annealing has the ability to give the best answer or if they are both equally good for the mentioned state-space graph with the given initial state. Include your reasoning. [2]

5. Consider the following game tree. The children of each node must be visited from left to right.

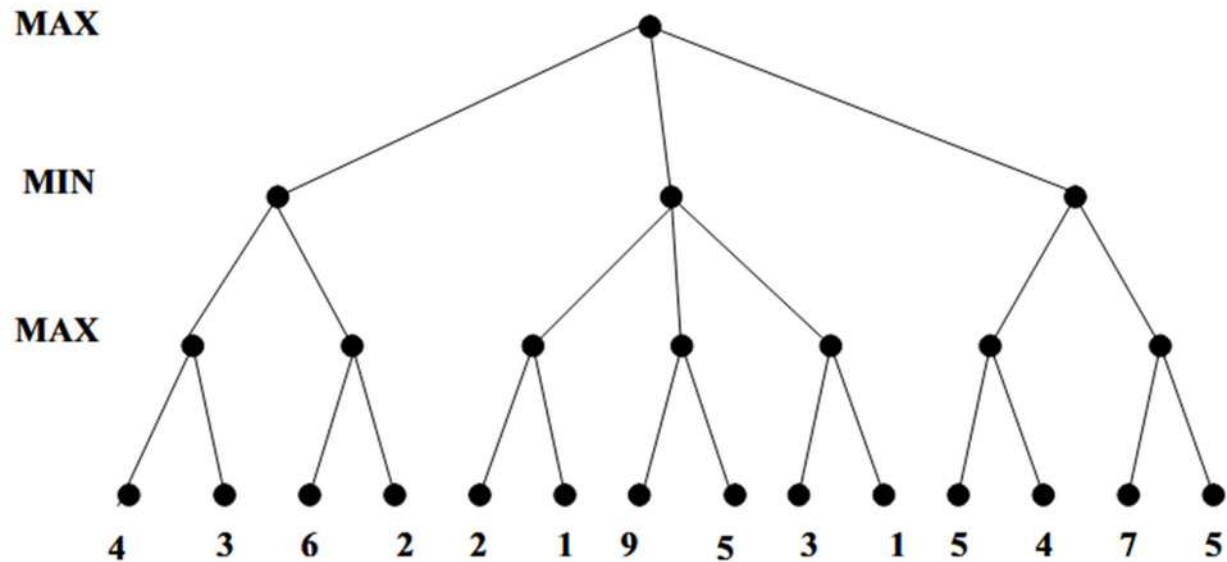


Figure 5(a)

a) Apply the MinMax algorithm with alpha-beta pruning to show which nodes will be pruned. Show utility[α , β , v] values for each node. [left to right] [4]

b) Does MinMax algorithm always provide optimal solution? If not give an example of such case. [2]